COMPETENCY STANDARD
FOR
PRODUCTION PROCESS IN WOVEN FABRIC MANUFACTURING
(TEXTILE SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance
Government of the People's Republic of Bangladesh

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The Competency Standard for **Production Process in Woven Fabric Manufacturing** is designed to guide the development of curricula, teaching materials, and assessment tools. It also serves as the document for providing training consistent with the requirements of the industry, so that individuals who pass through the set standard via assessment would be qualified and settled for a relevant job.
This document has been developed to be used in training under the **Skills for Industry Competitiveness and Innovation Program (SICIP)** to meet the skills requirements of the Textile sector. This document is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades, which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The textile sector is the largest manufacturing and export-earning sector of Bangladesh, and fabric manufacturing is the critical mid-stream stage that links yarn production with wet processing and garment manufacturing. Both the woven and the knit segments of the industry depend on this stage for the dimensional accuracy, structural quality and defect-free surface of the fabric that is finally converted into apparel. Weaving preparatory operations, weaving on shuttle and shuttleless looms, circular knitting and flatbed knitting are labor-intensive, machine-paced operations in which operator skill directly determines machine efficiency, fabric quality, waste percentage and the cost competitiveness of the entire downstream chain. This Competency Standard has therefore been developed to prepare a competent entry-level workforce for the production process in fabric manufacturing.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and to ensure that the required skills are aligned with industry requirements.
The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competency and its corresponding Elements.

OVERVIEW:

A Competency Standard is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

1. provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
2. enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies.
3. encourages the development and delivery of flexible training that suits individual and industry requirements.
4. encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standards are developed by a working group comprised of national process experts and the participation of experts from the industry to identify the competencies required of an occupation in a particular sector.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at a training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

1. Reference to Industry Sector, Occupational Title and Occupational Description
2. Unit code
3. Unit title
4. Unit descriptor
5. Unit of Competency
6. Elements and performance criteria
7. Variables and range statement
8. Evidence guides

Together all the parts of a Unit of Competency:

1. Describe a work activity
2. Guide the assessor in determining whether the candidate is competent.

This Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competency:

1. An overview of all Units of Competency for the occupation and their corresponding duration required for completion of training.
2. The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

UNITS & ELEMENTS AT A GLANCE:

Generic Competencies (20 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-TEX-PPFM-01-G	Apply Occupational Health and Safety (OHS) Practices in the Workplace	1. Identify OHS policies and procedures 2. Apply personal health and safety practices 3. Report hazards and risks 4. Respond to emergencies	10
SICIP-TEX-PPFM-02-G	Work in a Team Environment	1. Identify team goals and processes 2. Communicate and cooperate with team members 3. Work as a team member 4. Solve problems as a team member	10
Total Hours			**20 Hrs.**

Sector Specific Competencies (20 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-TEX-PPFM-01-S	Work in the Textile Sector	1. Identify the textile sector and its main activities 2. Understand basic roles in textile production processes 3. Recognize common tools and equipment used 4. Understand simple quality requirements in textile work	10
SICIP-TEX-PPFM-02-S	Apply Green Practices in the Textile Sector	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
Total Hours			**20 Hrs.**

Occupation Specific Competencies (320 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-TEX-PPFM-01-O	Interpret Yarns, Fabrics and Woven Fabric Manufacturing Processes	1. Identify yarns and raw materials used in fabric manufacturing 2. Identify woven fabric types, structures and end uses 3. Identify machines, tools and equipment used in woven fabric manufacturing 4. Describe the woven fabric manufacturing process sequence 5. Interpret work instructions and production documents	40
SICIP-TEX-PPFM-02-O	Perform Winding and Warping Operations	1. Prepare for winding and warping operations 2. Carry out yarn winding operations 3. Carry out warping operations 4. Monitor package and beam quality, and record production data	50
SICIP-TEX-PPFM-03-O	Perform Sizing Operations	1. Prepare for sizing operations 2. Carry out the sizing operation 3. Check sized beam and warp quality, and maintain records	60
SICIP-TEX-PPFM-04-O	Perform Weaving Operations	1. Prepare the loom for operation 2. Perform drawing-in, denting, knotting and knott-pass operations 3. Carry out the weaving operation 4. Attend warp and weft breakages and loom stoppages 5. Monitor fabric quality and control loom faults 6. Handle waste, cleaning and shift records	130
SICIP-TEX-PPFM-05-O	Perform Fabric Inspection and Grading	1. Prepare for fabric inspection 2. Carry out fabric inspection 3. Analyze fabric defects 4. Apply the fabric grading system 5. Report inspection results and take corrective action	40
Total Hours			**320 Hrs.**

COMPETENCY STANDARD: PRODUCTION PROCESS IN WOVEN FABRIC MANUFACTURING

The Generic Competencies

Unit of Competency: **APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES IN THE WORKPLACE**	**Nominal Duration:** 10 Hrs.	**Unit Code:** SICIP-TEX-PPYM-01-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply Occupational Health and Safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures; applying personal health and safety practices; reporting hazards and risks; and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1. 1 **OHS policies** and safe operating procedures are read and understood. 1. 2 Safety signs and symbols are identified and followed. 1. 3 Emergency response, evacuation procedures and other contingency measures are determined.
2. Apply personal health and safety practices	2. 1 OHS policies and procedures are followed and practiced. 2. 2 **Personal Protective Equipment (PPE)** is selected and used. 2. 3 Personal hygiene is maintained.
3. Report hazards and risks	3.1 **Hazards and risks** are identified, assessed and controlled. 3.2 Incidents arising from hazards and risks are reported to the authority. 3.3 Corrective actions are implemented to correct unsafe conditions in the workplace.
4. Respond to emergencies	4. 1 Alarms and warning devices are responded to. 4. 2 **Emergency response plans and procedures** are implemented. 4. 3 **First aid procedure** is applied during emergency situations.

Range of Variables:

Variable	Range (May include but not limited to)
1. OHS policies	1. 1 International OHS requirements 1. 2 Bangladesh standards for OHS 1. 3 Bangladesh Labour Act and Rules 1. 4 Building Code 1. 5 Fire Safety Rules and Regulations 1. 6 Industry guidelines and mill safety manual
2. Personal Protective Equipment (PPE)	2. 1 Nose/face mask (dust mask) 2. 2 Ear plugs or ear muffs 2. 3 Hand gloves 2. 4 Safety shoes 2. 5 Head cap/helmet 2. 6 Apron and overalls 2. 7 Goggles and safety glasses 2. 8 Hair net
3. Hazards and risks	3.1 Physical hazards (moving machine parts, nip points, noise, vibration, heat) 3.2 Chemical hazards (lubricants, cleaning agents, solvents) 3.3 Biological hazards (bacteria, fungus in damp areas) 3.4 Ergonomic hazards (repetitive work, standing, manual lifting) 3.5 Electrical hazards 3.6 Fire hazards (cotton dust, fly accumulation, static electricity)
4. Emergency response plans and procedures	4.1 Firefighting procedures 4.2 Earthquake response procedures 4.3 Evacuation procedures 4.4 Medical and first aid 4.5 Electrical accident response 4.6 Emergency machine stop procedures
5. First aid procedure	5. 1 Washing of open wound 5. 2 Washing chemically infected area 5. 3 Applying bandage 5. 4 Tourniquet 5. 5 Applying CPR (Cardiopulmonary Resuscitation) 5. 6 Taking appropriate medicine

Curricular Content Guide:

1. Underpinning Knowledge	1.1 OHS workplace policies and procedures. 1.2 Work safety procedures in a textile mill. 1.3 Emergency procedures. 1.3.1 Firefighting 1.3.2 Earthquake response. 1.3.3 Explosion response. 1.3.4 Accident response.

	1.4 Types of hazards (biological, chemical, physical, ergonomic and electrical) and their effects. 1.5 PPE types and uses. 1.6 Personal hygiene practices. 1.7 Safety signs, symbols and colour codes. 1.8 OHS awareness and reporting requirements.
2. Underpinning Skills	2. 1 Identifying OHS policies and procedures 2. 2 Following personal work safety practices 2. 3 Selecting and using appropriate PPE 2. 4 Reporting hazards and risks 2. 5 Responding to emergency procedures 2. 6 Maintaining physical well-being in the workplace 2. 7 Performing basic firefighting using fire extinguishers 2. 8 Applying basic first aid procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 PPEs 4.3 Firefighting equipment 4.4 Emergency response manual 4.5 Safety signs and symbols 4.6 First aid kits

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 followed OHS policies and procedures. 1.2 selected and used personal protective equipment (PPE). 1.3 reported incidents arising from hazards and risks to the authority. 1.4 implemented emergency response plans and procedures. 1.5 applied basic first aid procedure.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration

	2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **WORK IN A TEAM ENVIRONMENT**	**Nominal Duration:** 10 Hrs.	**Unit Code:** SICIP-TEX-PPYM-02-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to work in a team environment. It specifically includes the tasks of identifying team goals and processes, communicating and cooperating with team members, working as a team member and solving problems as a team member.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and processes	1.1 Team goals and collaborative decision-making processes are identified. 1.2 Roles and responsibilities of team members are identified. 1.3 Relationships within the team and with other workers are identified.
2. Communicate and cooperate with team members	2. 1 Effective interpersonal skills are used to interact with team members and to contribute to activities and objectives. 2. 2 Formal and informal **forms of communication** are used effectively to support team achievement. 2. 3 Diversity in character is respected and valued in team functioning. 2. 4 Views and opinions of other team members are understood and valued. 2. 5 Workplace terminology is used correctly to assist communication.
3. Work as a team member	3.1 Duties, responsibilities, authorities, objectives and task requirements are identified and clarified with the team. 3.2 Tasks are performed in accordance with organizational and team requirements, specifications, and workplace procedures. 3.3 Support to other team members is provided to ensure the team achieves its goals. 3.4 Agreed reporting lines are followed using standard operating procedures.
4. Solve problems as a team member	4.1 Current and potential problems faced by the team are identified. 4.2 A solution to the problem is identified. 4.3 Problems are solved effectively and the outcome of the implemented solution is evaluated.

Range of Variables:

Variable	Range (May include but not limited to)
1. Forms of communication	1. 1 Agenda 1. 2 Simple reports such as progress and incident reports 1. 3 Job sheets and shift handover log 1. 4 Operational manuals and SOPs 1. 5 Brochures and promotional material 1. 6 Visual and graphic materials 1. 7 Standards 1. 8 OSH information 1. 9 Signs and symbols

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Team goals and collaborative decision-making processes 1. 2 Roles and responsibilities of team members 1. 3 Relationships within the team and with other workers 1. 4 Effective interpersonal skills to interact with team members 1. 5 Effective formal and informal forms of communication 1. 6 Value of diversity in team functioning 1. 7 Correct use of workplace terminology 1. 8 Team's duties, responsibilities, authorities, objectives and task requirements 1. 9 Support mechanisms for other members of the team to ensure achievement of goals 1. 10 Methods of identifying current and potential problems faced by a team 1. 11 Effective problem-solving methods and evaluation of outcomes
2. Underpinning Skills	2. 1 Identifying team goals and collaborative decision-making processes 2. 2 Identifying roles and responsibilities of team members 2. 3 Using effective interpersonal skills to interact with team members 2. 4 Using formal and informal forms of communication 2. 5 Understanding and valuing views and opinions of other team members 2. 6 Performing tasks in accordance with organizational and team requirements, specifications and workplace procedures 2. 7 Supporting other members of the team to ensure the team achieves its goals 2. 8 Identifying current and potential problems faced by the team 2. 9 Identifying solutions to problems 2. 10 Solving problems effectively and evaluating the outcome of the implemented solution

3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Manuals/Guidelines 4.3 Pens 4.4 Papers 4.5 Workbooks

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 identified team goals and work processes. 1. 2 communicated and cooperated with team members. 1. 3 worked as a team member. 1. 4 solved problems as a team member.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 4. Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: **WORK IN THE TEXTILE SECTOR**	**Nominal Duration:** 10 Hrs.	**Unit Code:** SICIP-TEX-PPYM-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for work in the textile sector. It specifically includes the tasks of identifying the textile sector and its main activities, understanding basic roles in textile production processes, recognizing common tools and equipment used, and understanding simple quality requirements in textile work.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify the textile sector and its main activities	1. 1 The textile sector is identified within the context of national and global industries. 1. 2 **Main activities** of the textile sector are recognized. 1. 3 **Different stages of the textile value chain** are described in simple terms. 1. 4 Key workplaces in the textile sector are identified. 1. 5 Importance of the textile sector in the economy and the environment is understood.
2. Understand basic roles in textile production processes	2. 1 **Basic job roles in textile production** are identified and described. 2. 2 Simple production workflows and task sequences are understood. 2. 3 Teamwork and coordination in production processes are identified. 2. 4 Workplace safety roles and responsibilities are understood and followed. 2. 5 Shift working practices and reporting lines are understood.
3. Recognize common tools and equipment used	3.1 **Common tools used in textile processing** are identified correctly. 3.2 **Basic machines and equipment used in production** are recognized. 3.3 Functions of tools and equipment are understood in simple terms. 3.4 Basic maintenance and cleaning needs of tools and equipment are recognized.
4. Understand simple quality requirements in textile work	4.1 Basic quality standards in textile work are identified. 4.2 **Simple defects in textile products** are recognized. 4.3 Importance of maintaining quality during production is understood. 4.4 Work output is checked against basic quality

Elements of Competency	Performance Criteria
	expectations. 4.5 Good workmanship practices are followed to ensure quality.

Range of Variables:

Variable	Range (May include but not limited to)
1. Main activities	1. 1 Fibre production and preparation 1. 2 Yarn manufacturing (spinning) 1. 3 Fabric manufacturing (weaving and knitting) 1. 4 Wet processing (dyeing, printing and finishing) 1. 5 Garment manufacturing 1. 6 Testing and quality assurance 1. 7 Packaging, storage and export
2. Different stages of the textile value chain	2. 1 Natural and man-made fibre production 2. 2 Fibre preparation and mixing 2. 3 Yarn manufacturing (spinning) 2. 4 Fabric manufacturing (weaving/knitting) 2. 5 Dyeing, printing and finishing 2. 6 Cutting, sewing and finishing of garments 2. 7 Packaging, distribution and export
3. Basic job roles in textile production	3.1 Blow room operator/helper 3.2 Card tenter 3.3 Draw frame and comber operator 3.4 Simplex (speed frame) operator 3.5 Ring frame sider and doffer 3.6 Winding and packing operator 3.7 Quality checker/laboratory assistant 3.8 Maintenance fitter/electrician 3.9 Line leader/supervisor
4. Common tools used in textile processing	4.1 Sliver and roving cutter, scissors and nippers 4.2 Hooks, brushes and dusting tools 4.3 Spanner, screwdriver and plier set 4.4 Oil can and grease gun 4.5 Feeler and setting gauges 4.6 Wrap reel, wrap block and electronic balance 4.7 Tachometer and hand lens 4.8 Vacuum cleaner and air blowing gun
5. Basic machines and equipment used in production	5. 1 Bale opener/plucker and blow room line 5. 2 Carding machine 5. 3 Draw frame (breaker and finisher) 5. 4 Lap former and comber 5. 5 Speed frame (simplex) 5. 6 Ring frame 5. 7 Winding machine/autoconer

Variable	Range (May include but not limited to)
	5. 8 Humidification and waste collection plant 5. 9 Testing laboratory equipment
6. Simple defects in textile products	6. 1 Neps and slubs 6. 2 Thick and thin places 6. 3 Count variation and uneven yarn 6. 4 High hairiness 6. 5 Foreign fibre and contamination 6. 6 Sliver and roving breakage 6. 7 Soft/hard and damaged packages 6. 8 Oil and dirt stains

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Structure and importance of the textile sector in the national and global economy. 1. 2 Main activities and stages of the textile value chain. 1. 3 Position of yarn manufacturing within the textile value chain. 1. 4 Basic job roles, responsibilities and reporting lines in textile production. 1. 5 Simple production workflows, task sequences and shift practices. 1. 6 Common tools, machines and equipment used in textile production and their functions. 1. 7 Basic maintenance and cleaning needs of tools and equipment. 1. 8 Basic quality standards and common defects in textile products. 1. 9 Importance of quality, productivity and good workmanship in the workplace.
2. Underpinning Skills	2. 1 Identifying the textile sector and its main activities 2. 2 Describing the stages of the textile value chain in simple terms 2. 3 Understanding basic job roles and responsibilities in textile production processes 2. 4 Following simple production workflows, task sequences and teamwork practices 2. 5 Applying workplace safety roles and responsibilities in production activities 2. 6 Recognizing common tools, machines and equipment and their functions 2. 7 Recognizing basic maintenance and cleaning needs of tools and equipment 2. 8 Recognizing simple defects and checking work output against basic quality requirements 2. 9 Following good workmanship practices to ensure

	consistent quality output
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Charts, models and samples of fibre, sliver, roving and yarn 4.3 Multimedia projector, laptop and audio-visual clips 4.4 Learning manual and handouts 4.5 Stationery

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1. 1 identified the textile sector and its main activities. 1. 2 understood basic roles in textile production processes. 1. 3 recognized common tools and equipment used. 1. 4 understood simple quality requirements in textile work.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **APPLY GREEN PRACTICES IN THE TEXTILE SECTOR**	**Nominal Duration:** 10 Hrs.	**Unit Code:** SICIP-TEX-PPYM-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply green practices in the textile sector. It specifically includes the tasks of interpreting green concepts, minimizing resource use in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1. 1 Basic concepts of environmental sustainability and green practices are explained clearly. 1. 2 **Environmental impacts of textile production** processes are identified. 1. 3 **Resource efficiency concepts** are interpreted. 1. 4 Waste reduction, reuse and recycling principles are described in relation to yarn manufacturing operations. 1. 5 **Eco-friendly practices in the textile sector** are recognized and related to workplace activities.
2. Minimize resource use in the workplace	2. 1 Electricity, water, compressed air and raw materials are used efficiently according to workplace requirements. 2. 2 **Unnecessary consumption of resources** is identified and avoided during operations. 2. 3 Machines and equipment are operated in energy-efficient modes where applicable. 2. 4 Material wastage is minimized through careful handling and accurate measurement. 2. 5 **Resource-saving practices** are applied where possible. 2. 6 Leakage, spillage and wastage of materials are promptly controlled and reported. 2. 7 **Workplace practices** for resource conservation are followed.
3. Implement waste management practices	3.1 **Types of waste** generated in yarn manufacturing operations are identified correctly. 3.2 Waste materials are segregated according to workplace and environmental requirements. 3.3 **Reusable and recyclable materials** are separated from non-recyclable waste. 3.4 Waste is stored safely in designated containers or areas. 3.5 Waste disposal is carried out following environmental rules and organizational procedures. 3.6 Hazardous waste is handled carefully according to

Elements of Competency	Performance Criteria
	safety guidelines. 3.7 Waste reduction practices are applied to minimize environmental impact during operations.

Range of Variables:

Variable	Range (May include but not limited to)
1. Environmental impacts of textile production	1. 1 High electrical energy consumption of spinning machinery and humidification plants 1. 2 Cotton dust, fibre fly and micro-fibre emission 1. 3 Noise pollution from machines and compressors 1. 4 Solid waste generation (droppings, comber noil, soft and hard waste) 1. 5 Water consumption and effluent from wet processing 1. 6 Air emission and greenhouse gases from boilers and generators 1. 7 Used oil, grease and contaminated cotton waste
2. Resource efficiency concepts	2. 1 Energy-efficient motors and inverter/variable frequency drives 2. 2 Control of compressed air leakage and optimized air pressure 2. 3 Optimized humidification and ventilation control 2. 4 Reduction of soft and hard waste percentage 2. 5 Preventive maintenance to reduce power consumption 2. 6 Energy-efficient lighting and switching-off practices 2. 7 Optimum machine loading and reduction of idle running
3. Eco-friendly practices in the textile sector	3.1 Recycling of soft waste back into mixing where permitted 3.2 Use of recycled, organic or sustainably sourced fibre 3.3 Dust collection, filter and waste recovery systems 3.4 Compliance with Department of Environment (DoE) rules and regulations 3.5 Effluent treatment plant (ETP) and rainwater harvesting 3.6 Reuse of cans, bobbins, cones, cartons and packaging materials 3.7 Green building and energy audit practices
4. Unnecessary consumption of resources	4.1 Idle running of machines and motors 4.2 Compressed air leakage in pipelines and guns 4.3 Excess lighting in unoccupied areas 4.4 Over-cleaning with air guns instead of vacuum 4.5 Excess lubricant and cleaning material use 4.6 Excessive good fibre loss with droppings and noil
5. Resource-saving practices	5. 1 Switching off machines, lights and fans when not required 5. 2 Reporting air, water and oil leakage promptly

Variable	Range (May include but not limited to)
	5. 3 Using vacuum cleaning instead of air blowing where possible 5. 4 Reusing cans, tubes, cones and packaging materials 5. 5 Maintaining correct machine settings to reduce waste 5. 6 Recording and monitoring waste and energy use
6. Workplace practices	6. 1 Standard operating procedures (SOPs) 6. 2 Environmental policy of the organization 6. 3 Housekeeping and 5S practices 6. 4 Energy and waste monitoring records 6. 5 Instructions from supervisors and safety notices
7. Types of waste	7.1 Blow room droppings and card droppings/flat strips 7.2 Comber noil 7.3 Sliver, roving and pneumafil soft waste 7.4 Hard waste from winding (yarn and thread waste) 7.5 Damaged cans, bobbins, cones and tubes 7.6 Packaging waste (poly bags, bale wrappers, straps, cartons) 7.7 Used oil, grease and oil-contaminated cotton (hazardous waste) 7.8 Used filters, sweeps and floor waste
8. Reusable and recyclable materials	8.1 Usable soft waste for back-mixing 8.2 Comber noil for coarse count and open-end spinning 8.3 Cans, bobbins, tubes and cones 8.4 Cartons, poly bags and bale straps 8.5 Metal and plastic scrap 8.6 Paper and stationery waste

Curricular Content Guide:

1. Underpinning Knowledge	1. 1 Basic concepts of environmental sustainability, green practices and climate-resilient production. 1. 2 Environmental impacts of textile and yarn manufacturing operations. 1. 3 Resource efficiency concepts relating to energy, water, compressed air and raw material. 1. 4 Principles of waste reduction, reuse and recycling in a spinning mill. 1. 5 Eco-friendly practices and technologies applied in the textile sector. 1. 6 Types of waste generated at each stage of yarn manufacturing and permissible waste levels. 1. 7 Waste segregation, storage, disposal methods and compliance with environmental regulations. 1. 8 Hazardous waste handling and workplace environmental safety procedures. 1. 9 Relevant national environmental rules and organizational

	environmental policy.
2. Underpinning Skills	2. 1 Interpreting green concepts and relating them to workplace activities 2. 2 Identifying environmental impacts of yarn manufacturing operations 2. 3 Applying resource efficiency principles in workplace operations 2. 4 Using electricity, compressed air, water and raw materials efficiently 2. 5 Identifying and avoiding unnecessary consumption of resources 2. 6 Operating machines and equipment in energy-efficient modes where applicable 2. 7 Handling materials carefully to reduce wastage and improve accuracy 2. 8 Controlling and reporting leakage, spillage and resource wastage promptly 2. 9 Identifying and segregating different types of waste generated in yarn manufacturing 2. 10 Separating reusable and recyclable materials from non-recyclable waste 2. 11 Handling hazardous waste according to safety guidelines and regulations
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4. 1 Workplace (simulated or actual) 4. 2 Standard operating procedures 4. 3 Workplace documents, signs and symbols 4. 4 Codes of conduct and environmental policy 4. 5 Waste segregation bins and storage area 4. 6 Tools, equipment and facilities appropriate to the process or activities 4. 7 Materials relevant to the proposed activity

Assessment Evidence Guide:

1. Critical Aspects of Competency	assessment requires evidence that the candidate: 1. 1 interpreted green concepts. 1. 2 minimized resource use in the workplace.

	1. 3 implemented waste management practices.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: **INTERPRET YARNS, FABRICS AND WOVEN FABRIC MANUFACTURING PROCESSES**	**Nominal Duration:** 40 Hrs.	**Unit Code:** SICIP-TEX-PPFM-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to interpret yarns, fabrics and woven fabric manufacturing processes. It specifically includes the tasks of identifying yarns and raw materials used in fabric manufacturing, identifying woven fabric types, structures and end uses, identifying machines, tools and equipment used in woven fabric manufacturing, describing the woven fabric manufacturing process sequence, and interpreting work instructions and production documents.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify yarns and raw materials used in fabric manufacturing	1.1 **Types of yarn** used for weaving are identified according to fibre, structure and spinning system. 1.2 **Yarn count systems** are interpreted and simple conversions are performed. 1.3 Yarn quality parameters affecting weaving performance are recognized. 1.4 **Yarn package forms and identification documents** are interpreted.
2. Identify woven fabric types, structures and end uses	2.1 **Woven fabric types** are identified by examining samples, and the warp and weft directions, selvedge and surface features of each type are described. 2.2 **Woven fabric structures** are identified and distinguished from the weave pattern, using fabric particulars such as ends and picks per inch, GSM and fabric width. 2.3 **End uses** of each woven fabric are matched with its type, structure and properties.
3. Identify machines, tools and equipment used in woven fabric manufacturing	3.1 **Machines used in woven fabric manufacturing** are identified. 3.2 **Hand tools and measuring instruments** are identified and their functions are described. 3.3 Safety requirements for using machines, tools and equipment are identified.
4. Describe the woven fabric manufacturing process sequence	4.1 **Stages of the weaving process** are described in the correct sequence. 4.2 The purpose and output of each process stage are explained and linked to the next stage in the sequence. 4.3 Material flow and **production documents** are interpreted to

Elements of Competency	Performance Criteria
	trace a fabric from yarn input to inspected fabric.
5. Interpret work instructions and production documents	5.1 Production documents are identified and interpreted. 5.2 Work orders, fabric specification sheets and machine setting sheets are read and understood. 5.3 Standard operating procedures (SOPs) are followed as instructed. 5.4 Workplace terminology is used correctly during communication. 5.5 Information is recorded and reported using workplace formats.

Range of Variables:

Variable	Range (May include but not limited to)
1. Types of yarn	1.1 Carded and combed yarn 1.2 Blended yarn (cotton/polyester, cotton/viscose, cotton/lycra) 1.3 Ring spun 1.4 rotor/open-end 1.5 Single and ply (doubled) yarn 1.6 Filament yarn (POY, FDY, DTY) 1.7 Core-spun and elastane (spandex) covered yarn 1.8 Slub and fancy yarn
2. Yarn count systems	2.1 Indirect systems 2.1.1 English count (Ne) 2.2 Direct systems 2.2.1 Tex 2.2.2 Denier 2.3 Twist per inch (TPI) 2.4 Twist direction (S and Z)
3. Yarn package forms and identification documents	3.1 Cone, cheese, ring cop and pirn 3.2 Hank and warper's/weaver's beam
4. Woven fabric types	4. 1 Shirting 4. 2 Suiting 4. 3 Denim 4. 4 Twill/gabardine 4. 5 Poplin 4. 6 Canvas 4. 7 Sheeting 4. 8 Towel (terry) fabric 4. 9 Home textile fabric
5. Woven fabric	5.1 Plain weave

Variable	Range (May include but not limited to)
structures	5.2 Twill weave 5.3 Satin and sateen weave 5.4 Dobby 5.5 Jacquard 5.6 Pile weave
6. End uses	6.1 Shirts and blouses 6.2 Trousers and jeans 6.3 Uniforms 6.4 Bed linen 6.5 Towels 6.6 Curtains and upholstery 6.7 Bags and industrial or technical products
7. Machines used in woven fabric manufacturing	7.1 Winding machine 7.2 Warping machine 7.3 Sizing machine 7.4 Drawing-in frame and denting arrangement 7.5 Knotting machine 7.6 Shuttle loom (power loom) 7.7 Shuttle less looms/modern loom (rapier, air-jet, water-jet, projectile) 7.8 Fabric inspection machine
8. Hand tools and measuring instruments	8.1 Reed hook, drawing hook and denting hook 8.2 Scissors, nippers and thread cutter 8.3 Oil can and grease gun 8.4 Pick glass/counting glass and magnifier 8.5 GSM cutter and electronic balance
9. Stages of the weaving process	9.1 Yarn receiving, checking and conditioning 9.2 Winding 9.3 Warping 9.4 Sizing 9.5 Drawing-in, denting and knotting/tying-in 9.6 Weft preparation (pirn winding, where applicable) 9.7 Weaving on the loom 9.8 Fabric inspection and grading 9.9 Mending and spot cleaning 9.10 Folding, rolling, packing and dispatch
10. Production documents	10.1 Process flow chart 10.2 Route card 10.3 Work order 10.4 Production plan 10.5 Machine setting sheet

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Types of yarn used in weaving, their manufacturing systems and characteristics. 1.2 Yarn count systems, count conversion, twist, twist direction and twist multiplier. 1.3 Yarn quality parameters and their influence on weaving performance. 1.4 Classification of fabrics by method of manufacture. 1.5 Basic woven structures, their construction, appearance and identification. 1.6 Machines used at each stage of fabric manufacturing and their functions. 1.7 Hand tools and measuring instruments used in the weaving departments. 1.8 Sequence and objectives of the weaving process. 1.9 Process parameters and their influence on fabric quality and productivity. 1.10 Production documents, machine setting sheets, SOPs and workplace terminology.
2. Underpinning Skills	2.1 Identifying yarn types by appearance and simple physical examination 2.2 Determining yarn count using wrap reel and electronic balance 2.3 Identifying woven fabric structures using a pick glass 2.4 Interpreting fabric specification sheets and design sheets 2.5 Determining face and back sides and direction of fabric 2.6 Identifying machines, tools and measuring instruments used in fabric manufacturing 2.7 Describing the weaving process sequence and the objective of each stage 2.8 Identifying in-process material forms and tracing material flow 2.9 Interpreting work orders, machine setting sheets and production documents 2.10 Recording and reporting information using workplace formats
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual)

	4.2 Yarn sample board with yarns of different counts, types and packages 4.3 Fabric sample board with woven structures 4.4 Charts, models and flow diagrams of the weaving process 4.5 Wrap reel, wrap block and electronic balance 4.6 GSM cutter, GSM balance and pick glass 4.7 Hand tools and measuring instruments 4.8 Multimedia projector, laptop and audio-visual clips 4.9 Learning manual, SOPs and sample production formats

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 identified yarns and raw materials used in fabric manufacturing. 1.2 identified woven fabric types, structures and end uses. 1.3 identified machines, tools and equipment used in woven fabric manufacturing. 1.4 described the woven fabric manufacturing process sequence. 1.5 interpreted work instructions and production documents.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM WINDING AND WARPING OPERATIONS**	**Nominal Duration:** 50 Hrs.	**Unit Code:** SICIP-TEX-PPFM-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform winding and warping operations. It specifically includes the tasks of preparing for winding and warping operations, carrying out yarn winding operations, carrying out warping operations, and monitoring package and beam quality and recording production data.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for winding and warping operations	1.1 Work order, yarn details and lot information are obtained and understood. 1.2 Machines, creel and work area are checked for cleanliness and readiness. 1.3 **Safety devices and guards** are checked before start-up. 1.4 Required materials and accessories are arranged at the workstation. 1.5 Abnormal conditions found during checking are reported to the authority.
2. Carry out yarn winding operations	2.1 Yarn packages are creeled and threaded through the **yarn path** correctly. 2.2 **Winding parameters** are set according to yarn type and package requirement. 2.3 Yarn clearer settings are checked and applied as instructed. 2.4 **Winding faults** are identified and corrected during operation. 2.5 Package doffing is carried out and packages are placed in designated trolleys. 2.6 Yarn waste is collected and segregated as per workplace procedure.
3. Carry out warping operations	3.1 **Warping method** is applied according to the work order. 3.2 Creel is loaded and yarn ends are drawn through tensioners, guides and stop motions. 3.3 Warping parameters are set as per the warp plan. 3.4 Number of ends, beam width and warp length are maintained according to specification. 3.5 Colour and end sequence are maintained according to the sectional warping plan. 3.6 Broken ends are attended and joined promptly using the standard knotting method. 3.7 **Warping faults** are identified and corrected or reported. 3.8 Beam doffing, leasing, marking and identification are

Elements of Competency	Performance Criteria
	completed correctly.
4. Monitor package and beam quality, and record production data	4.1 Package and beam quality parameters are checked at specified intervals. 4.2 Yarn tension and beam density are verified against the standard. 4.3 **Common faults** are identified and reported to the authority. 4.4 Waste percentage is checked against the standard. 4.5 Production and quality data are recorded in the **prescribed format** as per the industry requirements. 4.6 Machine, creel and work area are cleaned after completion of work.

Range of Variables:

Variable	Range (May include but not limited to)
1. Safety devices and guards	1.1 Emergency stop buttons and pull cords 1.2 Machine guards, covers and interlock switches 1.3 Creel guards and yarn stop motions 1.4 Beam flange guard and doffing arrangement 1.5 Earthing and electrical safety devices 1.6 Warning signs and floor markings 1.7 Lockout-tagout arrangement
2. Yarn path	2.1 Package holder and unwinding accelerator/balloon breaker 2.2 Yarn guide and ceramic eyelet 2.3 Tensioner (disc, gate or capstan type) 2.4 Yarn clearer/slub catcher (mechanical or electronic) 2.5 Yarn detector and stop motion 2.6 Traverse drum and yarn guide finger
3. Winding parameters	3.1 Machine and speed parameters 3.2 Yarn tension 3.3 Yarn clearing (quality control)
4. Winding faults	4.1 Soft and hard package 4.2 Ribbon formation and patterning 4.3 Stitching and sloughing-off 4.4 Bad or weak splice/knot 4.5 Uneven package shape, bulged nose or base 4.6 Overfilled and underfilled package 4.7 Mixed count, mixed lot and shade mix 4.8 Oil, dirt and grease stain on package 4.9 Broken and entangled yarn on the package surface
5. Warping method	5.1 Direct (beam) warping for single-colour and large-lot production 5.2 Sectional (drum) warping for multi-colour and small-lot

Variable	Range (May include but not limited to)
	production
6. Warping faults	6.1 Missing, broken and lapped ends 6.2 Crossed ends and wrong end sequence 6.3 Wrong colour pattern and repeat error 6.4 Soft, hard or unevenly built beam 6.5 Entangled and snarled ends
7. Common faults	7.1 Soft or hard package and beam 7.2 Stitched or slough-off package 7.3 Ribbon formation 7.4 Bad splice and slipping knot 7.5 Missing, crossed and lapped ends 7.6 Uneven beam build and ridge formation 7.7 Oil, dirt and rust stains 7.8 Mixed count, mixed lot and shade variation 7.9 Excessive waste generation
8. Prescribed format	8.1 Winding production log sheet 8.2 Warping production log sheet 8.3 Beam and package card 8.4 Clearer cut and splice report 8.5 Machine stoppage and breakdown register 8.6 Quality checking sheet 8.7 Waste record 8.8 Shift handover log

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Types of winding machines, package forms and their applications. 1.2 Yarn path in a winding machine and the function of each element. 1.3 Winding parameters and their effect on package quality and unwinding performance. 1.4 Splicing and knotting methods, splice quality requirements and testing. 1.5 Winding faults, their causes, effects and remedies. 1.6 Creel types, creeling methods and yarn tensioning arrangements. 1.7 Warping parameters, warp plan and colour order interpretation. 1.8 Warping faults, their causes, effects and remedies. 1.9 Package and beam quality parameters and methods of checking. 1.10 Waste generation in winding and warping and permissible waste limits.

2. Underpinning Skills	2.1 Checking machines, creel and safety devices before start-up 2.2 Creeling yarn packages and threading the yarn path correctly 2.3 Setting winding parameters according to yarn type and package requirement 2.4 Performing splicing and knotting within the specified standard 2.5 Identifying and correcting winding faults during operation 2.6 Setting warping parameters as per the warp plan 2.7 Maintaining number of ends, beam width, warp length and colour sequence 2.8 Identifying and correcting warping faults 2.9 Doffing, leasing, marking and identifying beams 2.10 Checking package and beam quality parameters 2.11 Measuring yarn tension and beam density 2.12 Collecting, segregating and recording waste
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Cone/cheese winding machine and pirn winding machine 4.3 Direct (beam) warping machine and sectional warping machine 4.4 Creel with tensioners, guides and stop motions 4.5 Yarn packages of different counts and types 4.6 Splicer, knotter and knotting tools 4.7 Yarn tension meter and electronic balance 4.8 Wrap reel and wrap block 4.9 Trolleys, pallets and beam-handling arrangement 4.10 Learning manual, SOPs and sample production formats 4.11 Multimedia projector, laptop and audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	assessment requires evidence that the candidate: 1.1 prepared for winding and warping operations. 1.2 carried out yarn winding operations. 1.3 carried out warping operations. 1.4 monitored package and beam quality and recorded

	production data.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/ training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM SIZING OPERATION**	**Nominal Duration:** 60 Hrs.	**Unit Code:** SICIP-TEX-PPFM-03-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform sizing operations. It specifically includes the tasks of preparing for sizing operations, carrying out the sizing operation, and checking sized beam and warp quality, and maintaining records.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for sizing operations	1.1 Work order, warp plan and **size recipe** are obtained and understood. 1.2 Sizing machine, size box, creel and headstock are checked for cleanliness and readiness. 1.3 Sizing materials and chemicals are collected, weighed and checked for suitability. 1.4 **Safety devices and guards** are checked before start-up.
2. Carry out the sizing operation	2.1 Size paste is prepared in the **cooking vessel** following the recipe and the correct mixing sequence. 2.2 **Size solution parameters** are checked and maintained within the specified limit. 2.3 Warper's beams are creeled and the warp sheet is passed through the size box, drying zone and headstock. 2.4 **Sizing process parameters** are set and maintained as per the work order. 2.5 Size pick-up percentage, moisture content and stretch are maintained within the standard. 2.6 Splitting of the warp sheet and lease rod arrangement are maintained correctly. 2.7 Broken ends during sizing are attended and joined without disturbing the sheet. 2.8 Sized beam is doffed, marked and identified correctly.
3. Check sized beam and warp quality, and maintain records	4.1 Sized beam quality parameters are checked at specified intervals. 4.2 Size pick-up, moisture content and elongation are measured and recorded. 4.3 **Common sizing faults** are identified and reported to the authority. 4.4 Waste percentage is checked against the standard. 4.5 Production and quality data are recorded in the **prescribed format**. 4.6 Machine, size box, cooking vessel and work area are cleaned after completion of work.

Range of Variables:

Variable	Range (May include but not limited to)
1. Size recipe	1.1 Adhesive (native starch, modified starch, PVA, CMC, acrylic binder) 1.2 Softener and lubricant (wax, tallow, mineral oil) 1.3 Water quantity and total solid content 1.4 Separate recipes for cotton, polyester/cotton and filament warp
2. Safety devices and guards	2.1 Emergency stop buttons and pull cords 2.2 Guards over drying cylinders, squeeze rollers and headstock 2.3 Steam valve, pressure gauge and safety relief valve 2.4 Temperature and level alarms 2.5 Warning signs, floor markings and anti-slip flooring 2.6 Lockout-tagout arrangement
3. Cooking vessel	3.1 High-pressure cooker 3.2 Size mixing tank with stirrer 3.3 Size service/supply tank with jacket heating 3.4 Size box with immersion and squeeze rollers 3.5 Refractive index
4. Size solution parameters	4.1 Viscosity 4.2 Concentration and total solid content (Brix/refractive index) 4.3 Temperature of the paste in the cooker and size box 4.4 pH value 4.5 Cooking time and stirring speed 4.6 Level of size solution in the size box
5. Sizing process parameters	5.1 Machine (sizing) speed 5.2 Size box temperature 5.3 Squeeze roller pressure and number of nips 5.4 Immersion roller position and number of dips 5.5 Drying cylinder temperature and steam pressure 5.6 Size pick-up percentage 5.7 Beam pressure, beam density and winding tension 5.8 Splitting arrangement and lease rod setting
6. Common sizing faults	8.1 Over-sized (hard) and under-sized (soft) warp 8.2 Sticky ends and improper splitting 8.3 Uneven size pick-up across the sheet 8.4 Size lumps, shell and skin formation 8.5 Uneven moisture content and over-drying 8.6 Broken, crossed and lapped ends 8.7 Soft, ridged or unevenly built beam 8.8 Uneven beam edges and yarn falling over the flange

Variable	Range (May include but not limited to)
7. Prescribed format	9.1 Sizing production log sheet 9.2 Size recipe and cooking record 9.3 Size pick-up, moisture and stretch record 9.4 Beam card and set card 9.5 Drawing-in, denting and knotting job card 9.6 Machine stoppage and breakdown register 9.7 Waste record 9.8 Shift handover log

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Objectives and importance of sizing in the weaving preparatory process. 1.2 Construction and zones of a sizing machine (creel, size box, drying, headstock). 1.3 Sizing process parameters and their effect on warp quality and weaving performance. 1.4 Size pick-up, moisture content and stretch: definition, measurement and control. 1.5 Sizing faults, their causes, effects and remedies. 1.6 Hand knotting and machine tying-in methods. 1.7 Waste generation in sizing and permissible waste limits. 1.8 Safe handling and storage of sizing chemicals, steam and hot surfaces.
2. Underpinning Skills	2.1 Preparing size paste following the recipe and mixing sequence 2.2 Measuring and controlling viscosity, concentration, temperature and pH of size paste 2.3 Creeling warper's beams and passing the warp sheet through the machine 2.4 Setting and maintaining sizing process parameters 2.5 Measuring size pick-up, moisture content and stretch 2.6 Checking beam assembly for missing, crossed and double ends 2.7 Identifying sizing faults and their causes
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace

4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Sizing machine with size box, drying cylinders and headstock 4.3 Sizing chemicals and materials 4.4 Viscometer, refractometer/Brix meter, pH meter and thermometer 4.5 Moisture meter and electronic balance 4.6 Warper's beams and weaver's beams 4.7 Drawing-in frame with drop wires, healds, heald frames and reeds 4.8 Reed hook, drawing hook and denting hook 4.9 Warp tying/knotting machine 4.10 Waste segregation bins 4.11 Learning manual, SOPs, MSDS and sample production formats 4.12 Multimedia projector, laptop and audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 prepared for sizing operations. 1.2 carried out the sizing operation. 1.3 performed drawing-in, denting and knotting operations. 1.4 checked sized beam and warp quality and maintained records.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM WEAVING OPERATIONS**	**Nominal Duration:** 130 Hrs.	**Unit Code:** SICIP-TEX-PPFM-04-O

Unit Descriptor:
This unit covers the skills, knowledge and attitudes required to perform weaving operations. It specifically includes the tasks of preparing the loom for operation, performing drawing-in, denting, knotting and knott-pass operations, carrying out the weaving operation, attending warp and weft breakages and loom stoppages, monitoring fabric quality and controlling loom faults, and handling waste, cleaning and shift records.

Elements and Performance Criteria:
(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare the loom for operation	1.1 Work order, **fabric specification** and machine setting sheet are obtained and understood. 1.2 Loom plan is prepared based on ordered quantity and time. 1.3 Safety devices and guards are checked before start-up. 1.4 1.5 Sized beam, healds, reed, and drop wires are gaited and mounted correctly.
2. Perform drawing-in, denting, knotting and knott-pass operations	3.1 **Drawing-in accessories** are selected according to the fabric design. 3.2 **Drafting and denting plan** is interpreted from the design sheet. 3.3 Warp ends are drawn through drop wires, healds and reed in the correct order. 3.4 Denting is completed maintaining the specified ends per dent and selvedge arrangement. 3.5 Knotting/tying-in is performed as per job requirement. 3.6 New yarns along with knots are pulled through heald-eye and reed. 3.7 Weft package is mounted correctly. 3.8 Loom settings are verified against the machine setting sheet.
3. Carry out the weaving operation	2.1 **Primary, secondary and auxiliary motions** are observed for correct functioning. 2.2 Warp tension, pick density, ratio & insertion, and reed space are maintained as per specification. 2.3 Selvedge is made according to fabric finishing and yarn characteristics. 2.4 Colour, warp and weft pattern sequence are maintained according to the design.
4. Attend warp and weft	3.1 Warp breakages are attended and ends are joined using the

Elements of Competency	Performance Criteria
breakages and loom stoppages	weaving knot. 3.2 Weft breakage and weft exhaustion are attended and weft is replenished promptly. 3.3 Loom stop motions are attended and reset correctly. 3.4 Starting marks, thin places and thick places are prevented while restarting the loom. 3.5 Faults beyond the operator's scope are reported to the supervisor or mechanic.
5. Monitor fabric quality and control loom faults	4.1 Woven fabric is checked on the loom at specified intervals. 4.2 **Common woven fabric defects** are identified and their causes are determined. 4.3 Corrective actions within the operator's scope are taken to remove defects. 4.4 Fabric width, pick density and selvedge quality are verified against specification. 4.5 Defective fabric portions are marked and reported.
6. Handle waste, cleaning and shift records	5.1 Warp, weft and fabric waste are collected and segregated as per workplace procedure. 5.2 Waste percentage is maintained within the permissible limit. 5.3 Loom, under-loom area and accessories are cleaned as per the cleaning schedule. 5.4 Routine lubrication points are attended as per the schedule. 5.5 Production, efficiency and quality data are recorded in the **prescribed format** as per industry requirements. 5.6 Shift handover is completed with correct and complete information.

Range of Variables:

Variable	Range (May include but not limited to)
1. Fabric specification	1.1 Warp count and weft count 1.2 Ends per inch (EPI) and picks per inch (PPI) 1.3 Reed count, reed width and ends per dent 1.4 GSM 1.5 Warp and weft crimp and waste percentage 1.6 Selvedge type and selvedge width 1.7 Fabric length per roll and total order quantity
2. Drawing-in accessories	2.1 Drop wires (drop pins) - closed and open end 2.2 Heald wires, healds and heald frames/shafts 2.3 Reed of specified count and width 2.4 Lease rods and lease bands 2.5 Beam gaiting accessories and beam gears 2.6 Reed hook, drawing hook and denting hook

Variable	Range (May include but not limited to)
	2.7 Heald frame connecting rods and springs 2.8 Number of heald shafts according to the design repeat
3. Drafting and denting plan	3.1 Straight draft 3.2 Pointed and V draft 3.3 Skip and broken draft
4. Primary, secondary and auxiliary motions	4.1 Primary motions 4.1.1 Shedding 4.1.2 Picking (weft insertion) 4.1.3 Beat-up 4.2 Secondary motions 4.2.1 Take-up motion 4.2.2 Let-off motion 4.3 Auxiliary (tertiary) motions 4.3.1 Warp stop motion 4.3.2 Weft stop motion
5. Common woven fabric defects	5.1 Broken and missing ends and picks 5.2 Double end and double pick 5.3 Starting mark and stop mark 5.4 Reed mark and reed cut 5.5 Temple mark and temple cut 5.6 Float and wrong lifting 5.7 Weft bar and shade bar 5.8 Oil stain, dirt mark and rust mark 5.9 Loose, tight and cut selvedge 5.10 Hole, cut and tear 5.11 Snarl, slub and knot 5.12 Mispick and weft crack 5.13 Wrong draw and wrong denting 5.14 Uneven fabric width and uneven pick density
6. Prescribed format	6.1 Loom production and efficiency report 6.2 Pick counter reading sheet 6.3 Loom stoppage and breakdown register 6.4 Warp and weft breakage record 6.5 Fabric roll card and beam card 6.6 On-loom inspection sheet 6.7 Waste record 6.8 Lubrication and cleaning checklist 6.9 Shift handover log

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Classification of looms and the working principle of shuttle and shuttleless looms. 1.2 Loom settings and their effect on fabric quality and machine performance. 1.3 Purpose and methods of drawing-in, denting, knotting and gaiting. 1.4 Drop wires, healds, heald frames and reeds: types, specification and selection. 1.5 Drafting plans, lifting plans and denting plans and their interpretation. 1.6 Primary, secondary and auxiliary motions of the loom and their timing relationship. 1.7 Shedding devices: tappet, dobby and jacquard, and their selection for different designs. 1.8 Loom stop motions, their function and correct resetting. 1.9 Common woven fabric defects, their causes and remedies. 1.10 Waste generation in weaving and permissible waste limits. 1.11 Safe working procedures, machine guarding and lockout-tagout in the weaving shed. 1.12 Recording and reporting production, stoppage and quality data.
2. Underpinning Skills	2.1 Checking the loom, safety devices and guards before start-up 2.2 Mounting and threading the weft package and weft accumulator 2.3 Selecting drawing-in accessories according to the design 2.4 Drawing warp ends through drop wires, healds and reed 2.5 Denting with the specified ends per dent and selvedge arrangement 2.6 Operating tappet, dobby and jacquard shedding as per the design 2.7 Maintaining colour and weft pattern sequence 2.8 Identifying woven fabric defects on the loom and determining their causes 2.9 Doffing, marking and handling fabric rolls 2.10 Collecting, segregating and recording waste 2.11 Cleaning the loom and attending routine lubrication points 2.12 Recording production, efficiency, stoppage and quality data
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in

	the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Shuttle loom (modern/power loom) 4.3 Weft packages, pirns and weft accumulators 4.4 Machine setting sheets, design sheets and SOPs 4.5 GSM cutter and electronic balance 4.6 Tachometer and feeler gauges 4.7 Hand tool set and knotting tools 4.8 Oil can, grease gun and lubrication chart 4.9 Industrial vacuum cleaner and air blowing gun 4.10 Waste segregation bins 4.11 Learning manual and sample production formats 4.12 Multimedia projector, laptop and audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 prepared the loom for operation. 1.2 performed drawing-in, denting, knotting and knott-pass operations 1.3 carried out the weaving operation. 1.4 attended warp and weft breakages and loom stoppages. 1.5 monitored fabric quality and controlled loom faults. 1.6 handled waste, cleaning and shift records.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM FABRIC INSPECTION AND GRADING**	**Nominal Duration:** 40 Hrs.	**Unit Code:** SICIP-TEX-PPFM-05-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform fabric inspection and grading. It specifically includes the tasks of preparing for fabric inspection, carrying out fabric inspection, analyzing fabric defects, applying the fabric grading system, and reporting inspection results and taking corrective action.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare for fabric inspection	1.1 Work order, **fabric specification** and buyer requirements are obtained and understood. 1.2 Inspection machine, lighting arrangement and work area are checked for readiness. 1.3 Inspection tools and instruments are collected and checked for accuracy.
2. Carry out fabric inspection	2.1 Fabric is inspected on the face side at the specified speed and viewing distance. 2.2 Fabric parameters are measured and verified against the specification. 2.3 Defects are identified as per **inspection methods.** 2.4 Defects are marked with stickers, tags or flags at the selvedge.
3. Analyze fabric defects	3.1 **Fabric defects** are classified according to industry standard. 3.2 Major and minor defects are distinguished according to the standard. 3.3 Repairable defects are separated for mending.
4. Apply the fabric grading system	4.1 Fabric grading system is applied as per the buyer or organizational requirement. 4.2 Penalty points are assigned to defects according to defect length and severity. 4.3 Points per 100 square yards or square metres are calculated correctly. 4.4 Fabric is graded as acceptable or rejected against the standard limit. 4.5 Grade and inspection results are recorded roll-wise.
5. Report inspection results and take corrective action	5.1 Inspection results are recorded in the prescribed format as per industry requirement. 5.2 Inspection reports are prepared and submitted to the

Elements of Competency	Performance Criteria
	authority. 5.3 Mending and spot cleaning are arranged for repairable defects.

Range of Variables:

Variable	Range (May include but not limited to)
1. Fabric specification	1.1 Fabric construction = (Warp count x weft count / EPI x PPI) 1.2 Fabric width 1.3 Weave design 1.4 GSM
2. Inspection methods	2.1 4-point system 2.2 10-point system
3. Fabric defects	3.1 Starting mark 3.2 Fabric hole 3.3 Loose/tight/miss/double warp/weft 3.4 Thick/thin/neps/slub warp/weft 3.5 Oil and dirt spot 3.6 Reverse/cut/broken pick

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Fabric specification, buyer requirements and tolerance limits. 1.2 Construction and operation of the fabric inspection machine. 1.3 Identification of the face side and direction of woven and knitted fabric. 1.4 Measurement of fabric width, length, weight, GSM, EPI, PPI, WPI and CPI. 1.5 Skewness, bowing and spirality: causes and measurement. 1.6 Woven fabric defects, their appearance, causes and responsible process. 1.7 Fabric grading systems (4-point, 10-point, 2.5-point and Graniteville '78'). 1.8 Calculation of penalty points per 100 square yards and per 100 square metres. 1.9 Documentation, reporting and traceability requirements.
2. Underpinning Skills	2.1 Interpreting fabric specifications and buyer quality requirements 2.2 Checking the inspection machine, lighting and instruments for readiness 2.3 Identifying the face side and direction of the fabric 2.4 Inspecting fabric at the specified speed and viewing

	distance 2.5 Counting EPI, PPI, WPI and CPI using a pick glass 2.6 Checking shade variation against the approved swatch and shade band 2.7 Identifying and classifying woven fabric defects 2.8 Applying the 4-point and other grading systems 2.9 Segregating and identifying rejected and second-quality fabric
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Fabric inspection machine with light box 4.3 Colour matching cabinet with multiple light sources 4.4 Fabric samples showing a wide range of defects 4.5 GSM cutter, GSM balance and electronic balance 4.6 Measuring tape, steel rule and length counter 4.7 Fabric trolley and roll-handling arrangement 4.8 Inspection formats, grading sheets and buyer quality manuals 4.9 Learning manual, SOPs and reference standards 4.10 Multimedia projector, laptop and audio-visual clips

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 prepared for fabric inspection. 1.2 carried out fabric inspection. 1.3 identified fabric defects and determined their causes. 1.4 applied the fabric grading system. 1.5 reported inspection results and took corrective action.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training centre or in an actual or simulated workplace after completion of the training.

	3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

-END OF COMPETENCY STANDARD-

WORKSHOP / LAB FACILITY STANDARD

Course Name:	**Production Process in Woven Fabric Manufacturing**
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/Lab/Classroom cum Workshop):

1. Classroom - 350 sft (33 sqm)
2. Workshop/Lab - 1800 sft (167 sqm), **OR**
3. Classroom cum workshop - 2100 sft (195 sqm)

Major Training Equipment and Training Facilities:

S.N.	**Major Equipment and Training Facilities**	**Required Quantity**
1.	Computer/Laptop	1
2.	Multimedia projector & screen	1
3.	Whiteboard with accessories	1
4.	Winding machine	1
5.	Warping machine	1
6.	Sizing machine	1
7.	Modern Loom	10
8.	Fabric inspection machine	1

The following conditions must be fulfilled –

1. The institute shall not use the same facilities for any other projects/organizations offering a similar course.
2. The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facilities required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirements of the Competency Standard (CS) provided by SICIP.

For the operation of the training course on **Production Process in Woven Fabric Manufacturing**, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

Where a training institute does not possess the full-scale production machinery, the institute must establish a formal linkage with a weaving and/or knitting factory so that the practical sessions of the occupation-specific units can be conducted in an actual workplace under supervision.

The institute will also provide all other hand tools and power tools as per the CS for 25 trainees. The institute will also arrange adequate seating arrangements and classroom setup for the 25 trainees.